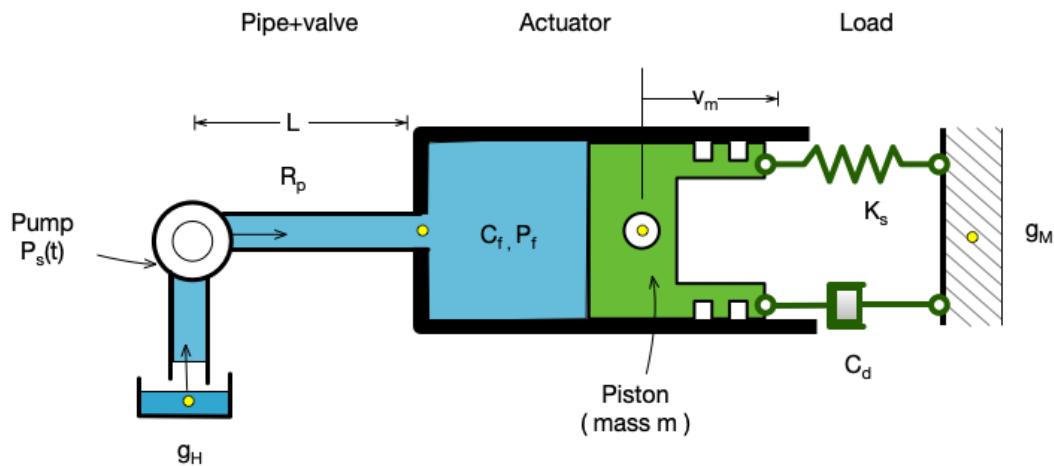


SURNAME NAME MAT. NUMBER

Signature _____

1 Modelling: Exercise 1

A pressure-controlled fluid actuator is used to move a mechanical load represented in the figure ?? with a spring stiffness K_s and damping coefficient C_d . The pump provides hydraulic fluid at pressure $P_s(t)$, which enters the actuator chamber through a pipe and a valve of fluid flow resistance R_p . The actuator chamber has a fluid capacitance C_f due to the **compressibility** of the fluid, whose pressure is P_f . The oil bulk modulus is β . This pressure moves the load whose mass is m and the area of cross-section is A , against the mechanical resistance of stiffness K_s and damping coefficient C_d . The mass is moved inside the actuator chamber at speed v_m . The pressure $P_s(t)$ is controlled.



Respond to the following questions:

- 1 Draw the corresponding linear graph and the normal tree.
- 2 write the incident matrix and derive the **circuit equations** and the **node equations**.
- 3 Write the list of **constitutive equations** for each element in the hydraulic-mechanical system.
- 4 Assemble the equations and specify the list of input variables, the list of system states and the list of equations of dynamics.

5

Report in the corresponding boxes the final result of what is required. All the calculation done to arrive to the final solution can be derived manually on give paper or using Matlab Symbolic Toolbox or Maple. The paper with manual detailed solution must be handed to the examiner and the Maple or Matlab file must be uploaded to the Moodle platform.

Draw the corresponding Linear Graph (LG) and the normal tree.:

Report the incidence matrix.

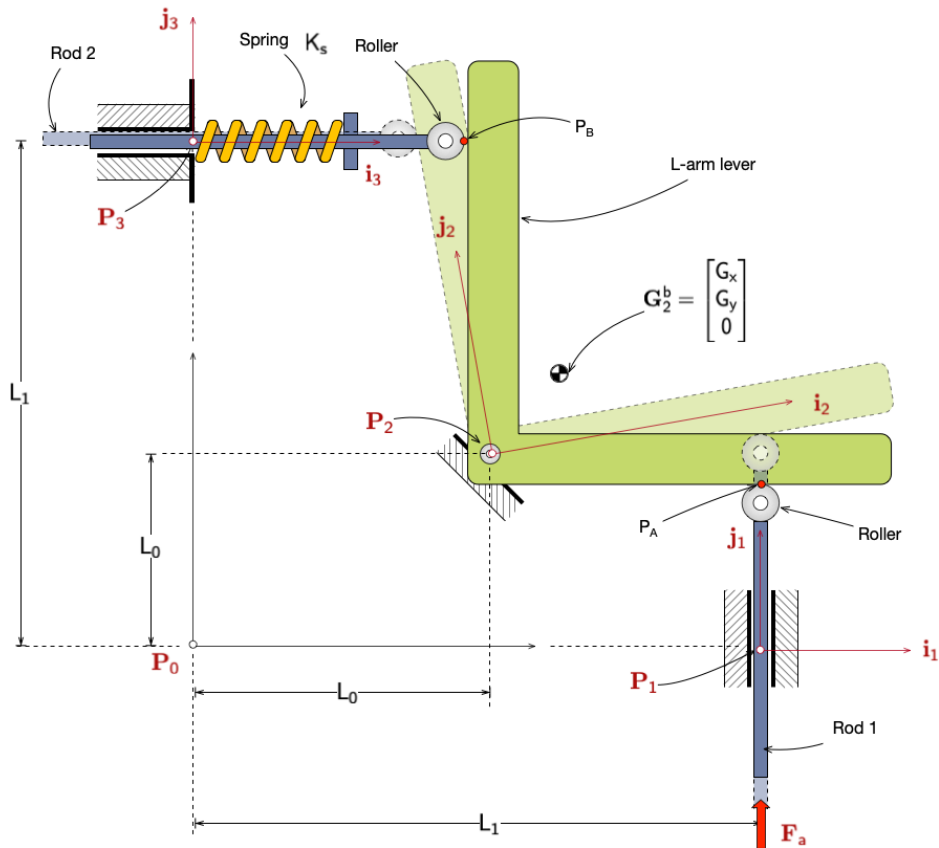
Report the final set of equations.

2 Modelling: Exercise 2

The figure ?? shows a spring loaded L-shaped lever arm. The mechanism is drawn in the initial configuration with the spring uncompressed and external force $\mathbf{F}_a$ equal to zero. The dimensions shown in the picture are in the world reference frame $\mathbf{i}_0, \mathbf{j}_0$ and can be used to derive all other necessary dimensions. When the external force $\mathbf{F}_a$ is applied to the rod 1, the arm rotates around the point $\mathbf{P}_2$ and pushes backward the rod 2 (according to positive sign of absolute x-axis $\mathbf{i}_0$) and the spring is compressed. Both rods are connected to the ground with $\mathbf{P}_1$ and $\mathbf{P}_3$, which correspond to the initial locations of points $\mathbf{P}_1$ and $\mathbf{P}_3$, respectively, the origins of the two rod bodies' frames $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{i}_3, \mathbf{j}_3$.

The L-shaped lever arm is constrained to rotate with respect to the ground around the point $\mathbf{P}_2$. The two rods are in contact with the L-shaped lever arm at the points $\mathbf{P}_A$ and $\mathbf{P}_B$ with two rollers. We can neglect the dimensions of the rollers and consider the contact points as points. Additionally the contact points are assumed to slide along the x - and y - axes of the L-arm lever without friction.

The L-shaped lever arm is the only body with the mass m with centre of mass in $\mathbf{G}_2$ and rotational inertia with respect to the $\mathbf{G}_2$ equal to I_z . The two rods are assumed to be massless and the spring is linear with stiffness K_s .



Respond to the following questions:

- 1 Draw the LG and the normal tree of the system
- 2 derive the kinematic constraint equations using the recursive formulation (i.e. loop equations)
- 3 derive the expression of the point $\mathbf{G}_2$, $\mathbf{P}_A$ and $\mathbf{P}_B$.
- 4 derive the node equations and the equations of motion using the Newton Euler approach

Suggestion: consider the rods always in contact with the L-shaped lever arm and also use the reference frames defined in the picture.

Report in the corresponding boxes the final result of what is required. All the calculation done to arrive to the final solution can be derived manually on give paper or using Matlab Symbolic Toolbox or Maple. The paper with manual detailed solution must be handed to the examiner and the Maple or Matlab file must be uploaded to the Moodle platform.

Draw the corresponding Linear Graph (LG) and the normal tree.:

Report the loop equations.

Report the node equations.

Report the final set of equations of motion.